

v_{c_r} be two nodes in $G(A(H))$ connected by a path $P = (v_{c_1}, \dots, v_{c_r})$ with $r \geq 2$. We add a new hyperplane h to $A(H)$. Let A_i and A_{i+1} be the corresponding arrangement before and after h is added, respectively. For each cell c_j in A_i , we denote the corresponding cell in A_{i+1} by $\bar{c}_j$. If $c_j \neq \bar{c}_j$, we denote the two connected cells that bisect $\bar{c}_j$ by c'_j and c''_j , respectively.

We show that there exists a $(v_{\bar{c}_1}, v_{\bar{c}_2})$ -path P' in $G(A_{i+1})$. First, let $c_1 = \bar{c}_1$. If $c_2 = \bar{c}_2$, we can choose $P' = (v_{c_1}, v_{c_2})$. If $c_2 \neq \bar{c}_2$, the connectivity of c'_2 and c''_2 ensures that we can choose $P' \in \{(v_{c_1}, v_{c'_2}), (v_{c_1}, v_{c''_2})\}$.

Second, let $c_1 \neq \bar{c}_1$. As c_1 and c_2 are connected in $G(A_i)$, it follows that c'_1 or c''_1 is connected to $\bar{c}_2$ in $G(A_{i+1})$. In the first case, we can choose $P' \in \{(v_{c'_1}, v_{c_2}), (v_{c'_1}, v_{c'_2}), (v_{c'_1}, v_{c''_2})\}$ depending on whether c_2 is divided by h or not. In the second case, we can replace c'_1 by c''_1 and obtain the sought path. Inductively, there exists a $(v_{\bar{c}_1}, v_{\bar{c}_r})$ -path P' in $G(A_{i+1})$ and the claim follows. $\square$

The graph connectivity ensures that, starting from any node $v_{c'}$, all nodes v_c in $G(A(H^=))$ can be reached using a breadth-first search. The choice of the incidence graph $G(A(H^=))$ prevents us from reaching nodes that do not belong to a cell of $A(H^=)$. The incidence graph $G(A(H^=))$ can be easily derived from the incidence lattice $G'(A(H^=))$ constructed in the algorithm of Edelsbrunner et al. (1986) to compute the arrangement $A(H^=)$. If two cells c_1 and c_2 share a common face f of dimension $p - 1$, there exists a node v'_f in $G'(A(H^=))$ and an edge from v'_f to each of the nodes v'_{c_1} and v'_{c_2} belonging to the cells c_1 and c_2 . In $G(A(H^=))$, there is an edge between v_{c_1} and v_{c_2} and we store the information to which separating hyperplane $h(e, f)$ this edge belongs. This requires constant time per face of dimension $p - 1$ and, thus, the graph $G(A(H^=))$ can be obtained in time linear in the number of faces of dimension $p - 1$ of $A(H^=)$ from $G'(A(H^=))$. Of course, we store the representative point from the relative interior of each cell. We are now ready to state our improved algorithm.

Theorem 3.6. Let $p \geq 2$. Algorithm 2 solves the p -parametric matroid problem in time $\mathcal{O}(m^{2p}f(m))$.